INCH-POUND

MIL-A-27195A(USAF)
10 August 1989
SUPERSEDING
MIL-A-27195(USAF)
27 May 1959

MILITARY SPECIFICATION

ACCELEROMETER, AIRCRAFT, MECHANICAL ABU-6/A

This specification is approved for use by all Departments of the Air Force and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of integrally lighted 1.875 inch dial, -2 to +8g range accelerometer designated ABU-6/A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Method Of
DOD-D-1000	Drawings, Engineering And Associated Lists
MIL-C-5541	Chemical Conversion Coatings On Aluminum And Aluminum Alloys

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEOR, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6610

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MIL-A-8625	Anodic Coatings For Aluminum And Aluminum Alloys
MIL-L-25467	Lighting, Integral, Red, Aircraft Instrument, General Specification For
MIL-P-26514	Polyurethane Foam, Rigid Or Flexible, For Packaging
MIL-L-27160	Lighting, White, Integral, Instrumental, General Specification For
MIL-C-83488	Coating, Aluminum, Ion Vapor Deposited

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-810C	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-838	Lubrication Of Military Equipment
MIL-STD-889	Dissimilar Metals
MIL-STD-970	Standards And Specification, Order Of Precedence For The Selection Of
DOD-STD-1866	Soldering Process General (Non Electrical)
MIL-STD-2073/-1	DOD Material Procedures For Development And Application Of Packaging Requirements
MS33558	Numerals And Letters, Aircraft, Instrument Dial, Standard Form Of
MS33585	Pointer, Dial, Standard Design Of Aircraft Instrument
MS33638	Case, Instrument, Flange Mounted, Aircraft

(Unless otherwise indicated, copies of federal and military specifications, and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia PA 19120-5099.)

(Copies of MIL-STD-810C are available from the Director, Naval Publications & Printing Service, 700 Robbins Ave, Bldg 4 Section D, Philadelphia, PA 19111.)

2.1.2 Other government documents, drawings, and publications. The following other Government dcuments, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Non-Government standards and other publications are normally available from the organization that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein (except for related associated detail specifications, specifications sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.4), a sample shall be subjected to first article inspection in accordance with 4.5.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-970 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.2.2 Standard parts. With the exception in 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 MATERIALS

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the contracting activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts except where magnetic materials are essential.

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3.3.3 Nonferrous materials. Nonferrous materials shall be used for all parts of the accelerometer except where ferrous materials are essential.

3.3.4 Metals. Metals shall be of the corrosin-resistant type or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

3.3.5 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.6 Corrosive fumes. The materials as installed in the accelerometer shall not liberate deleterious fumes.

3.3.7 Protective treatment. When materials are used in the construction of the accelerometer that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction. The accelerometer shall be designed to indicate aircraft vertical accelerations in maneuvers and confirm to figure 1; meet the response characteristics specified in Section 4; constructed to permit overhaul or repair without the use of special tools or fixtures.

3.4.1 Indication and range. The accelerometer shall indicate vertical acceleration when the plane of the mounting lugs is vertical and the +1 graduation is at 270°. When in this position, normal gravity shall be indicated at +1g. The accelerometer shall be designed to indicate accelerations in the range -2 to +8g units (See 6.3.2).

3.5 Performance. The accelerometer shall be capable of meeting the test requirements specified in section 4 when subjected to the following conditions:

a. Temperature. Temperature ranging from +71°C to -62°C. Storage temperature of -65°C.

b. Humidity. Relative humidity up to 100 percent.

c. Altitude. Pressures ranging from 30 inches Hg down to 1.31 inches Hg approximately 70,000 feet altitude.

d. Fungus. Fungus growth as encountered in tropical climates.

e. Salt fog. Exposure to salt sea atmosphere for a period of 50 hours.

f. Thermal shock. Thermal shock with alternate immersion in water at temperatures of +85°C and +5°C for 8 cycles.

g. Acceleration and shock. Acceleration and shock forces as anticipated.

h. Vibration error. When vibrated at an amplitude of 0.018 to 0.020 inch and at a frequency variation between 8.3 and 50Hz for a total of 3 hours the main pointer shall not exceed the 0.2g to 0.4g range and the auxiliary pointers shall not change more than 0.2g.

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i. Vibration endurance. The dynamic scale error shall not exceed 0.25g when subjected to the vibration endurance test.

j. Static scale error. Static scale error shall not exceed when tested as specified.

k. Dynamic scale error. Dynamic scale error shall not exceed ±0.25g at any specified test point.

l. Magnetic effect. With the accelerometer revolved about a magnetic compass with a magnetic field intensity of 0.17 to 0.19 oersted, the compass shall not deflect more than 1°.

m. Mounting lugs. The mounting lugs shall withstand a load of 175 pounds without fracture.

n. Response. Response characteristics shall be within the limits specified in 4.4.6, when the accelerometer is vibrated at frequencies from 4 to 20 Hz at maximum g forces for each test point.

3.5.1 Construction. The accelerometer shall be so constructed as to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service, without failure.

3.5.1.1 Pivots, bearings and gears. Pivots, bearings, and gears shall neither bind nor shake and shall be as near frictionless as practicable.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-5028 shall govern changes in the manufacturer's part numbers.

3.7 Dial. The dial shall conform to figure 2 and shall be fastened securely to the case or upon the frame of the mechanism by not less than 2 screws in such a manner that it will not loosen or turn when the instrument is vibrated.

3.7.1 Distance from dial to cover glass. Distance between dial and cover glass shall not be greater than 0.250 inch.

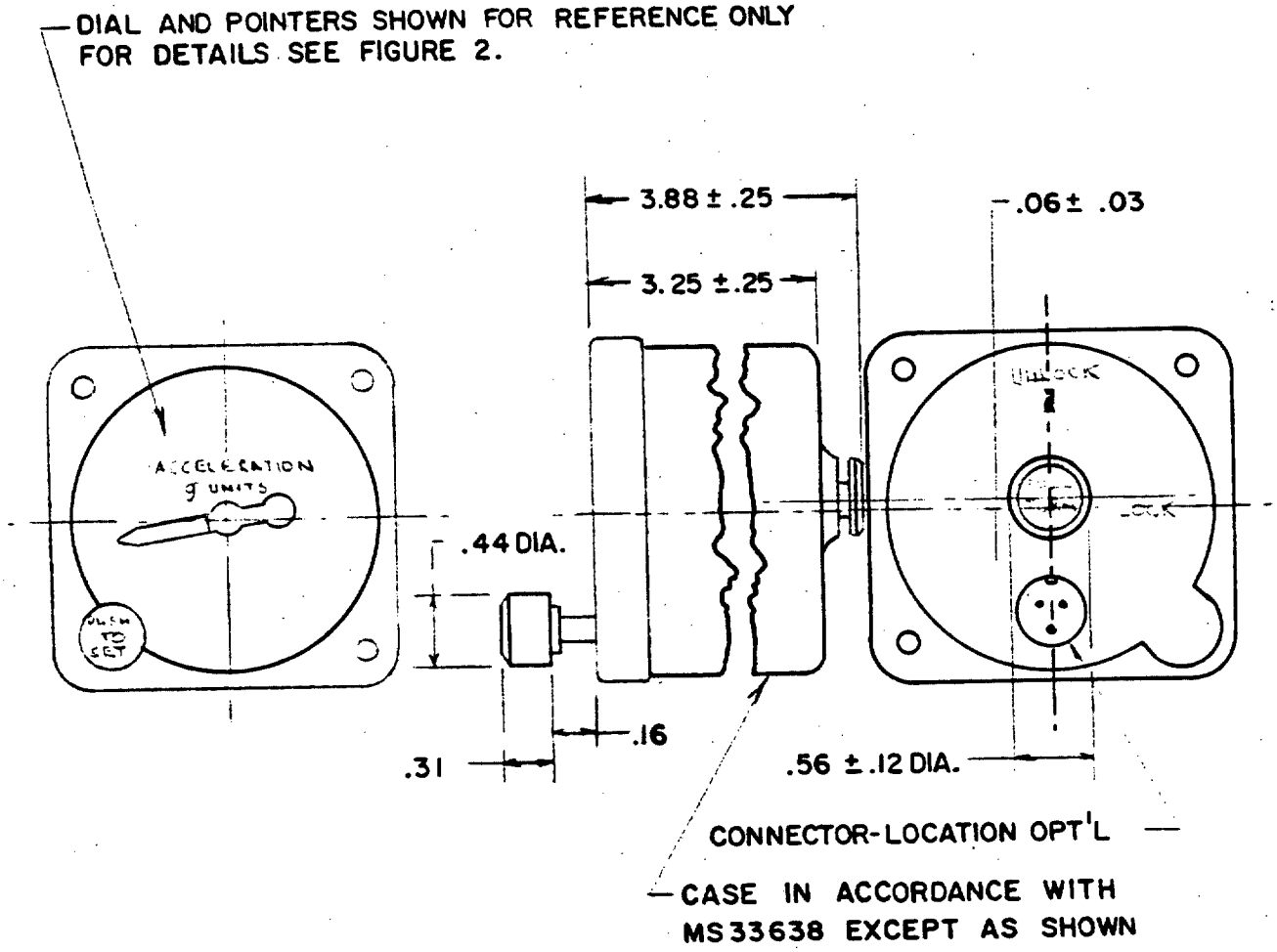
3.7.2 Numerals and letters. The style and proportion of numerals and letters, except the letter "g", placed on the dial of the accelerometer shall conform to MS33558. Numerals shall distinctly indicate the graduation to which each applies.

3.8 Pointers. The accelerometer shall have 3 pointers. One main pointer for indicating instantaneous acceleration and 2 auxiliary pointers; one for indicating maximum negative acceleration. The pointers shall conform to figure 2. They shall be light and sufficiently rigid to prevent oscillation under vibration. They shall be firmly attached to the mechanism, yet shall be readily adjustable.

3.8 1 Ratchet mechanism. Each auxiliary pointer shall be provided with a ratchet device, in order that indication of maximum plus and minor deflections of the main pointer will be maintained until the auxiliary pointers are reset to obtain a new reading.

3.8.2 Auxiliary pointers return mechanism. A mechanism shall be provided for resetting the auxiliary pointers into juxtaposition with the main pointer. A knob, located as shown on figure 1, shall be provided for operating the return mechanism.

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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES: ± .02

FIGURE 1. Case

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Stops shall be provided in order that the auxiliary pointers when reset shall not indicate a lower or higher reading that indicated by the main pointer. Stops shall be a part of the mechanism connecting the pointers and shall not directly engage them. Operation of the return mechanism shall require 2.50 to 8 pounds of force on the knob.

3.8.3 Main pointer mechanism. The main pointer mechanism shall not engage the mechanism of the auxiliary pointers when the imposed acceleration is less than the acceleration indicated by the auxiliary pointers.

3.9 ACCELERATION-RESPONSIVE ELEMENT

3.9.1 Stops for acceleration-responsive element. Mechanical stops shall be provided for arresting the motion of the acceleration-responsive element when the accelerometer is subjected to accelerations beyond the limits specified. One stop shall engage the element at an acceleration of not less than -2.1g and more than -2.2g, and the other at an acceleration of not less than +8.1g and not more than +8.2g.

3.9.2 Locking device. A device for manually locking the position of the acceleration-responsive element shall be provided, if the design is such as to require it in order to protect the mechanism of the accelerometer from possible damage during shipment. If provided, this device shall be controlled by means of a knurled knob located in the back of the case as shown on figure 1.

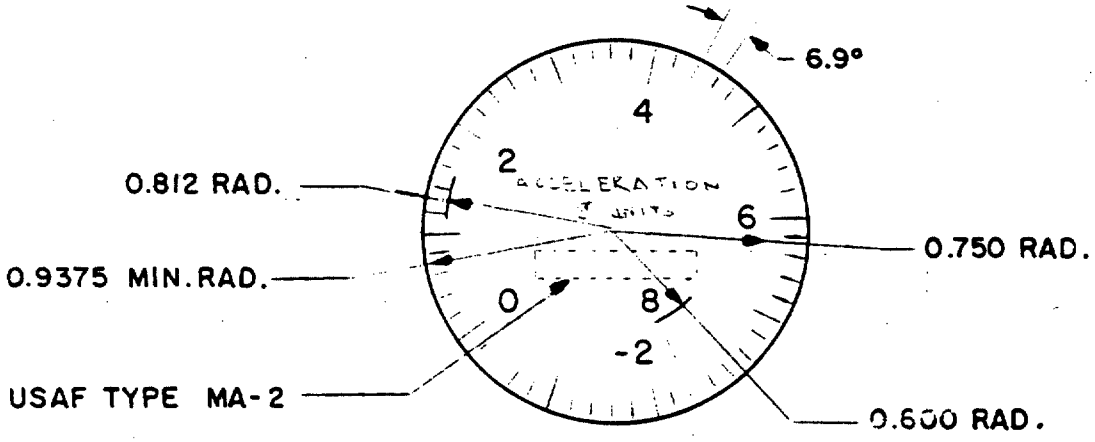
3.9.3 Adjustment. The mechanism shall be provided with means to adjust or correct the indications. This means of adjustment shall be reasonably simple to manipulate and of such nature that adjustment can be accomplished with tools ordinarily possessed by an instrument repairman.

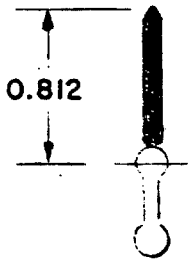
3.10 Integral lighting. The indicator shall be integrally lighted with 2 parallel lighting circuits, a white lighting circuit and a red lighting circuit, as shown on figure 3. The red lighting shall be in accordance with MIL-L-25467 and the white lighting shall be in accordance with MIL-L-27160. In the event there is any conflict between this specification and either MIL-L-25467 or MIL-L-27160, this specification shall govern. Although resistors shall not be permitted in the original design of the lighting systems, space shall be allocated for incorporation of a trimming resistor in each lighting circuit within the indicator. The magnitude of the trimming resistors shall be specified by the contracting activity in terms of the brightness decrease required. The lighting shall be subjected to approval by the contracting activity.

3.10.1 Testing, integral lighting. The integral lighting shall be comparable to and in brightness balance with a "prime standard" indicator. The prime standard shall be an indicator of the type described in this specification. The prime standard shall have applied to its lighting circuits the specified voltage which the contracting activity has established as being necessary to assure equality of presentation with respect to the lighting with adjacent but dissimilar instruments. The contractor shall submit the test procedure for determining compliance with this paragraph to the contracting activity for approval.

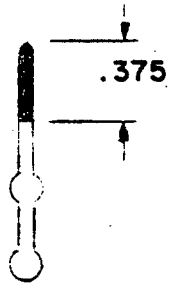
3.11 Case. The case shall conform to figure 1 and shall be finished in durable dull black. The material shall be suitable for hermetic sealing, uniform in texture, and which shall provide a smooth surface.

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MAIN POINTER
MS33585-1



AUXILIARY POINTER
MS33585-1
EXCEPT AS SHOWN

DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES: FRACTIONS ± 1/64
DECIMALS ± 0.010
ANGLES ± 0°10'

FIGURE 2. Dial and pointers.

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3.12 Marking. All markings shall be durable to withstand the usage encountered in service. Numerals shall distinctly indicate the graduation to which each applies. The following markings shall be finished with dull white material, color No 37038 of FED-STD-595. The dimensions of the markings, when applicable shall be as follows:

	Height or length Inch ±0.016	Width of line or graduation Inch ±0.005
Numerals	0.188	MS33558
Graduations corresponding to numerals	See figure 2	0.031
Lettering "g UNITS" and "ACCELERATION"	0.016	MS33558
One "g" graduation		0.062
Shaded part of pointers	———	———
Knob markings	———	———
Small graduations	See figure 2	0.015

3.12.1 Durable dull black. The following markings shall be durably and legibly marked on the dial. These markings, the background of the dial, and all other markings not otherwise, specified shall be finished in durable dull black.

	Height or length Inch - 0.016
Lettering "USAF" and "ABU-6/A	0.063

3.12.2 Visibility of dial. The pointers , numerals, at least 0.063 inch of the shortest graduations, and other specified markings on the dial shall be visible from any point within the frustrum of a cone whose side makes an angle of 30° with a perpendicular to the dial and whose small diameter is the aperture of the instrument case.

3.14 Soldering. Soldering shall be accomplished in accordance with MIL-STD-1866.

3.15 Lubrication. The accelerometer shall be lubricated in accordance with MIL-STD-838.

3.16 Weight. The weight of each accelerometer shall not exceed 1.75 pounds.

3.17 FINISHES AND PROTECTIVE COATINGS

3.17.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625 except as follows:

a. Dials, small holes, and case inserts need not be anodized.

b. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

c. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

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DEUTSCH CONNECTOR DM 5600-3P, OR EQUAL

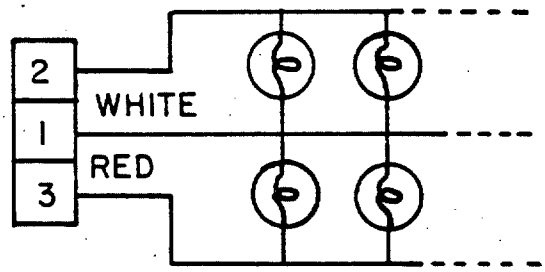


FIGURE 3. Lighting circuit

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d. Castings containing nonaluminum integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

e. When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.17.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practical, in accordance with MIL-C-83488, Type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coatings, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory with MIL-C-83488, may be used if demonstrated to be satisfactory and is approved by the cognizant engineering activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

3.18 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.19 WORKMANSHIP

3.19.1 General. The accelerometer, including all parts and accessories, shall be fabricated and finished in a thoroughly workman like manner in accordance with MIL-STD-454, requirement 9. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, and riveting; alignment of parts and tightness of assembly screws, and bolts.

3.19.2 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciable tightened further without damage or injury to the screw or bolt or threads.

3.19.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.19.4 Gears. Gear assemblies shall be properly aligned and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.19.5 Cleaning. The accelerometer shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

3.20 Envelope. An envelope, containing the mounting screws, shall be furnished with each indicator. The following information shall be printed on the face of the envelope:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS

3.21 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material either indicated or actual, nor does it commit the Government to accept defective material.

4.1.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Acceptance tests (see 4.2).

b. First article inspections (see 4.5)

4.2 Acceptance tests. Acceptance tests shall be consist of:

a. Individual tests

b. Sampling plans and tests

4.2.1 Individual tests. Each accelerometer shall be subjected to the following tests as described under 4.4 TEST METHODS:

a. Examination of product (see 4.4.1)

b. Static scale error and friction (see 4.4.2)

c. Dynamic scale error (see 4.4.3)

d. Response characteristics (4.4.5.1)

e. Return mechanism operation (see 4.4.7)

f. Internal lighting (see 4.4.4)

4.2.2 SAMPLING PLANS AND TESTS

4.2.2.1 Sampling plan A. One accelerometer selected at random from each 100 or less produced on the contract or order shall be subjected to the following tests as described under 4.4 TEST METHODS:

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a. Individual tests (see 4.2.1)

b. Low temperature operation (see 4.4.8)

c. High temperature operation (see 4.4.9)

d. Magnetic effect (see 4.4.10)

e. Vibration error (see 4.4.11)

f. Response characteristics (4.4.5.1)

4.2.2.2 Sampling plan B. Unless otherwise specified, 3 accelerometers selected at random from the first 15 items of the contract or order shall be subjected to the following tests as described under 4.4 TEST METHODS:

a. Sampling plan A tests (see 4.2.2.1)

b. Vibration endurance (see 4.4.12)

c. Shock (see 4.4.13)

d. Mounting lugs (see 4.4.14)

e. Low temperature-high altitude (4.4.15)

f. High temperature exposure (see 4.4.16)

g. Thermal shock (see 4.4.17)

h. Humidity (see 4.4.18)

i. Fungus (see 4.4.19)

j. Salt fog (see 4.4.20)

4.2.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no item still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.2.2.3.1 Individual test may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.2.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the contracting activity of all defects likely to be found and methods of correcting them.

4.3 Inspection conditions. Unless otherwise specified all inspections shall be performed in accordance with the test conditions specified in 4.3.1 of this specification.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that

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the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Vibration stand. A vibration stand shall be used which will vibrate at any desired frequency between 8.3 and 50 Hz and shall subject the accelerometer to vibration such than a point on the case will describe in a plane inclined 45° to the horizontal plane, a circle, the diameter of which is equal to the amplitude specified herein (see 6.3.1).

4.3.3 Calibration procedure. The accelerometer shall be calibrated based upon the vertical distance of the center of mass of the assembly from the center of the dial for each required test points. Each test point computed on the basis of a rotating arm whose radius to the center of the instrument dial is 9.375 inches. The formula used to compute the speed (rpm) required is as follows:

$$N = 9.5493 \sqrt{\frac{32.2 \text{ "g"}}{r}}$$

Where N = RPM of rotating
"g"= Test Point
r = Radius of arm in feet, corrected for mass position

4.3.3.1 Test points. "N" shall be computed for all points from -2g to +8g in increments of 1g.

4.3.4 Adjustments. No adjustments or repairs of any type shall be performed on the preproduction test samples after start of preproduction tests.

4.4 METHOD OF INSPECTION.

4.4.1 Examination of product. Each accelerometer shall be examined externally to determine conformance with this specification with respect to material, workmanship, marking, and envelope.

4.4.2 Static scale error. The position error shall be observed with accelerometer held in each of the following positions:

a. Plane of dial vertical with +1g graduation at 270°.

b. Plane of dial horizontal.

c. Plane of dial vertical with +1g graduation at 90°.

In each position, the main-pointer reading obtained shall be due entirely to gravity and not motion in change of position of instrument. The static scale error shall not exceed ±0.2g in any of the 3 positions tested.

4.4.3 Dynamic scale error. The accelerometer shall be mounted on a suitable centrifuge and centrifuge rotor per Air Force Drawing 8346561 and tested for dynamic scale error and friction for accelerations of -2g to 8g at 1g increments for a total of 10 test points. For the first test point and without instrument vibration, the centrifuge shall be brought up to, but not exceeding, the rotation rate equivalent to the "g" level being tested. The centrifuge rate shall be reduced to zero and the accelerometer reading taken from the memory pointer. After re-zeroing the memory

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pointer, the test shall be repeated except instrument vibration shall be applied utilizing an electric motor and an attached counterweight as defined by Air Force Drawing 8346561.

The centrifuge shall again be reduced to zero and the accelerometer reading taken from the memory pointer. Again, re-zero the memory pointer. The reading with and without vibration shall not differ from the centrifuge input "g" reading by more than ±.25g. This procedure shall be repeated for the remaining 9 test points. The centrifuge rate need not be returned to zero and the memory pointer reset, if capability exists to read the instruments without reducing the centrifuge's rotational rate.

4.4.4 Integral lighting. Each lighting circuit shall be properly connected individually and tested to show compliance with the requirements of 3.10.

4.4.5 Response characteristics (sampling plan A). The accelerometer shall be subjected to vibration along an axis parallel to the motion of the sensing element with the plane of the dial vertical and with the 1g graduation in the 9 o'clock position. The test apparatus used shall be as specified in 4.4.5.2. The accelerometer shall be tested at each of a minimum of 17 different frequencies. These tests shall be approximately equally spaced through the frequency range from 4 to 20 Hz. The amplitude of vibration at each test point shall be such that the maximum g forces practicable for that test point shall be obtained. The indicated g shall be recorded and a plot made for each test point of indicated g/ applied g vs frequency. The points so plotted shall fall within the shaded area on graph on figure 4. These graphs shall be a part of the preproduction test data.

4.4.5.1 Response characteristics (Individual tests). This test shall be identical to the test specified in 4.4.5 except that the accelerometer shall be tested at the following frequencies only: 4, 6, 8, 12, 16, and 20 Hz.

4.4.5.2 Test apparatus. The equipment shall impose linear sinusiodal motion to the accelerometer and shall be capable of procuring the frequencies and double amplitude required for each test point. The acceleration vector of any harmonic measured on the test table shall not exceed 10 percent of the fundamental acceleration vector.

4.4.7 Return mechanism operation. The accelerometer shall be placed in operating position with the pointer deflected to the maximum indication, the reset knob shall then be operated to return the pointer to zero. The force applied to the reset shall be between 2.5 and 8 pounds.

4.4.8 Low temperature operation. The accelerometer shall be subjected to the low temperature test, Method 502.1, Procedure I of MIL-STD-810C.

4.4.9 High temperature operation. The accelerometer shall be subjected to the high temperature test, Method 501.1, Procedure I of MIL-STD-810C for a period of 4 hours. Upon completion on the test period and while at the test temperatures, the accelerometer shall be subjected to and meet the individual tests.

4.4.10 Magnetic effect. The accelerometer shall be revolved about a short bar magnet compass with the nearest part of the indicator 5.50 inches from the bar magnet, and in a vertical plane which is a perpendicular bisector of the north-south axis of the bar magnet. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated on its horizontal axis until it is in its normal operating position. The horizontal

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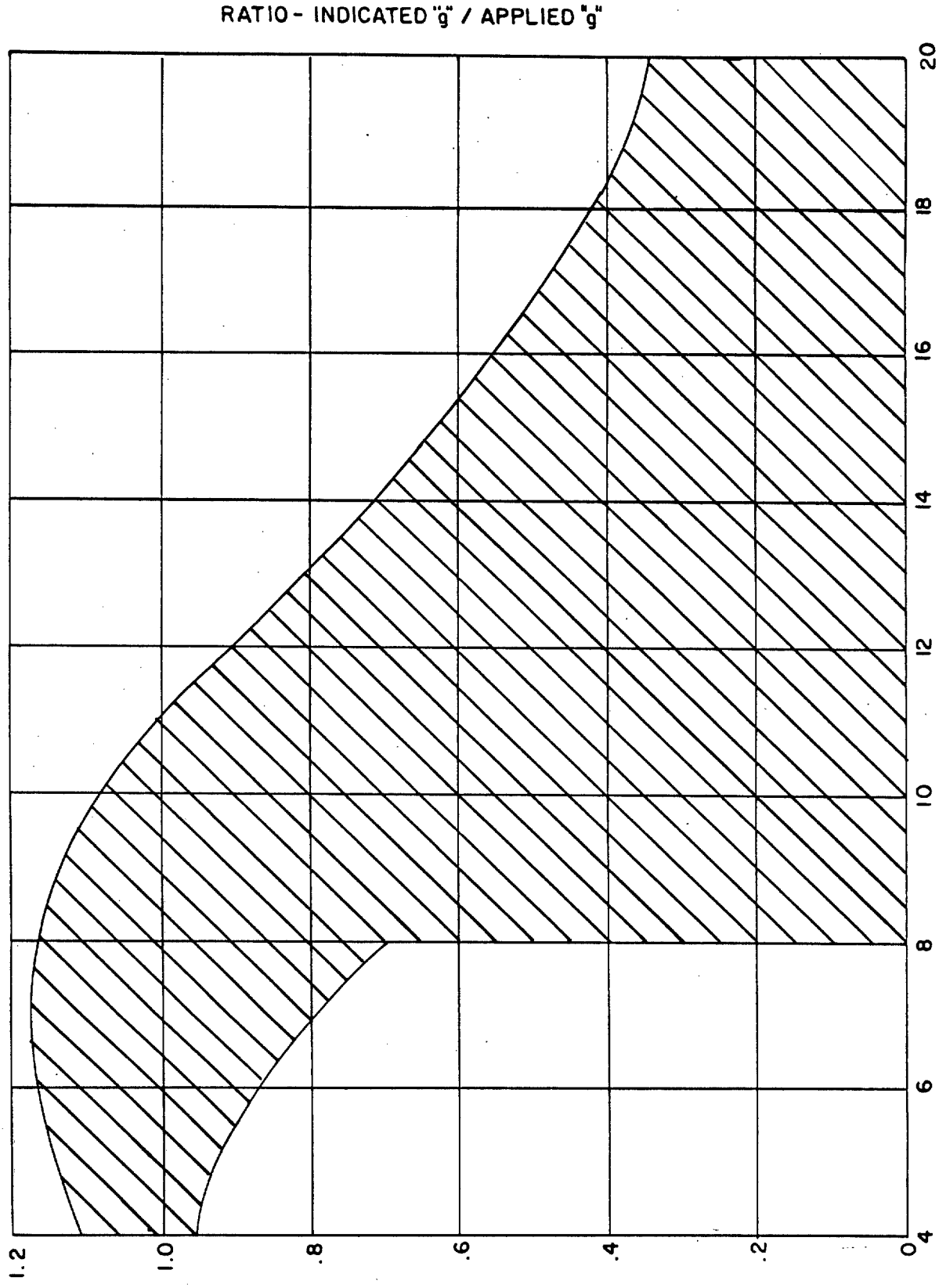


FIGURE 4. Frequency of applied acceleration (HZ).

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magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 1°.

4.4.11 Vibration error. The accelerometer shall be mounted on a suitable vibration stand with the dial in a vertical plane and with the graduation +1 at 270°. The auxiliary pointers shall be displaced approximately 1g from the main pointer and the readings noted. The reset mechanism shall not be operated until the completion of the tests. For a total of 3 hours the accelerometer shall be vibrated at the specified frequencies at an amplitude between 0.018 and 0.020 inch, and the pointer oscillation shall be noted. The frequency shall be varied from 8.3 to 50 Hz in a period of not more than 60 minutes. The natural frequency of the seismic system shall be noted during the frequency variation between 8.3 and 50 Hz. For the remainder of the 3 hour test period, the accelerometer shall be vibrated at a frequency of 20 Hz. At the end of the test period, the indication of the auxiliary pointer shall be read before operating the reset mechanism. At frequencies below 20 Hz the oscillation of the main pointer shall not exceed 0.2g and at frequencies over 20 Hz, the oscillation shall not exceed 0.3g. After vibration, the indications of the auxiliary pointers shall not be changed by more than 0.1g. The accelerometer shall not be damaged as a result of this test.

4.4.12 Vibration endurance. The instrument shall be mounted on a suitable vibration table in its normal operating position. The accelerometer shall be vibrated with circular motion of 0.018 to 0.020 inch diameter in a plane inclined 45 degrees to the horizontal plane and the frequency of vibration shall be varied uniformly from 5 to 50 cycles per second and return once each hour for a 50-hour period. Upon completion of this test period the accelerometer shall be subjected to the tests specified in 4.4.3. The dynamic scale error shall not exceed $\pm$0.25g.

4.4.13 Shock. The accelerometer shall be installed in the normal operating position in a free-fall apparatus as described in 4.4.13.1. If provided with a locking device, the accelerometer mechanism shall be locked. The accelerometer shall be allowed to fall from a height of 8 inches, then examined for damage and subjected to the test specified in 4.4.3. There shall be no damage to the accelerometer and the steady state scale error shall not exceed 0.25g.

4.4.13.1 Apparatus. A rigidly constructed device provided with a panel mounting for installing the accelerometer in the specified position, and vertical guides to allow free-fall from the specified height shall be used. A bumper on the free-fall element shall arrest its fall by striking the center of a rounded projection on a metal block. The bumper shall be spring steel, 0.25 by 1.50 inches in cross section with an unsupported length of 5 inches between points of attachment to the free-fall element. The bumper shall contact the metal block at the midpoint of a broad side. The center of gravity of the instrument under test and the contact point of the bumper shall be aligned vertically within 0.13 inch. The weight of the free-fall element shall be approximately 12 pounds. The metal block shall weigh not less than 50 pounds.

4.4.14 Mounting lugs. The accelerometer case with mechanism and cover glass removed shall be mounted face downward on the movable head of a suitable testing machine with the diameter of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine. A load of 175 pounds shall be applied along the axis of the mounting hole and toward the front of the case for a period of 1 minute. There shall be no fracture.

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4.4.15 Low temperature-high altitude. The accelerometer shall be subjected to the low temperature test, Method 502.1, Procedure I of MIL-STD-810C, and an altitude of 70,000 feet (1.31 inches Hg); except the temperature shall be held at the lowest operating temperature for a period of 72 hours. There shall be no damage to the accelerometer as a result of this test.

4.4.16 High temperature exposure. The accelerometer shall be subjected to the high temperature test, Method 501.1 of MIL-STD-810C. No damage to the accelerometer shall result from this test.

4.4.17 Thermal shock cycling. The accelerometer shall be alternately subjected to a total of 8 cycles of immersion in tap water maintained at $85^{\circ} \pm 2^{\circ}C$ and $5^{\circ} \pm 2^{\circ}C$. The length of time for each immersion shall be 30 minutes; not more than 5 seconds shall elapse between immersions. No evidence of moisture penetration or damage to the inclosure shall result from this test. Following this test, the accelerometer shall be subjected to and meet the requirements for the test specified in 4.4.4.

4.4.18 Humidity. The accelerometer shall be subjected to the humidity test in accordance with Method 507.1, Procedure I, of MIL-STD-810C, except the temperature shall be $71^{\circ}C$ ($160^{\circ}F$) in step 2.

4.4.19 Fungus. The accelerometer shall be subjected to the fungus resistance test in accordance with Method 508.1 of MIL-STD-810C.

4.4.20 Salt fog. The accelerometer shall be subjected to the salt fog test in accordance with Method 509.1 Procedure I for a period of 50 hours. Upon completion of the 50 hour test period, of MIL-STD-810C there shall be no evidence of external corrosion or discoloration as a result of this test.

4.5 First article inspection

4.5.1 First article inspection sample tested by the contractor. The contractor shall subject 3 accelerometers to the first article inspection.

4.5.2 First article inspection. The first article inspection shall consist of all the test specified under 4.4 TEST METHODS.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Accelerometers shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Accelerometers shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each accelerometer shall be packaged in quantity unit packs of one each in accordance with Method I-C1 of MIL-P-116. Each accelerometer shall be placed in a

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PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the accelerometer when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each accelerometer shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The industrial preservation of accelerometers shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2)

5.2.1 Level A. Accelerometers packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Accelerometers packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged accelerometer shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but not mandatory.)

6.1 Intended use. The ABU-6/A accelerometer covered by this specification is intended for use on aircraft to indicate vertical acceleration in maneuvers and in rough air.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Level of packaging and packing desired

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d. Whether sampling plan B tests are to be conducted

e. Reinspection date

6.3 DEFINITIONS

6.3.1 Amplitude. Whenever the word AMPLITUDE is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.3.2 "g" UNIT. The "g" unit is defined as an acceleration of 32.2 feet per second per second.

6.4 First article. When a first article is required, the item will be inspected and tested in accordance with Section 4 of this document. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

6.5 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement NAT-STD-3330 and AIR STD 10/4. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

6.6 Subject term (key words) listing.
Intergal Lighted
Vertical accelerometer

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
AIR FORCE -99

Preparing activity:
AIR FORCE -71

Project No:
6610-F273

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-27195A(USAF)	Accelerometer, Aircraft ABU-6/A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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